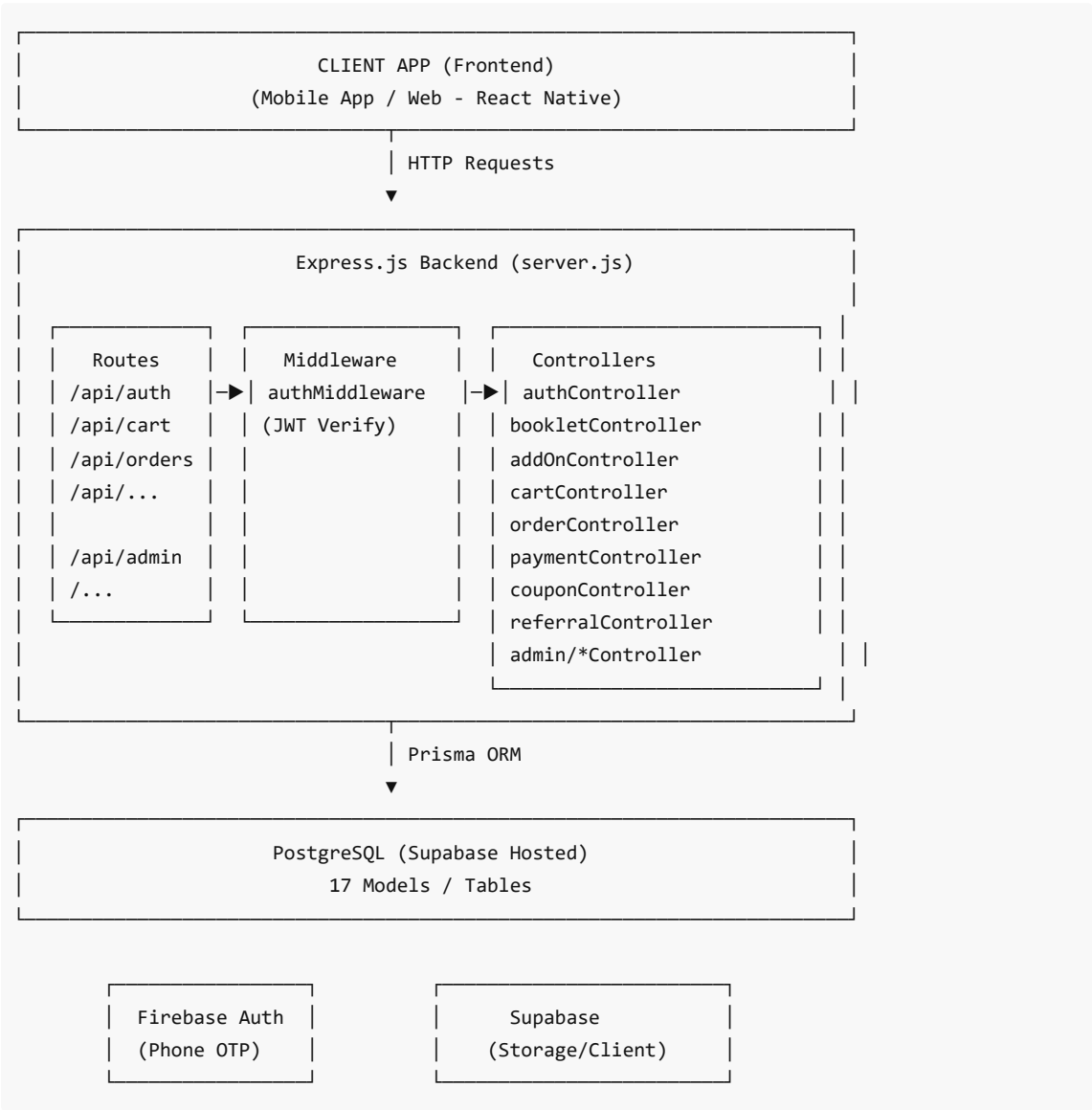
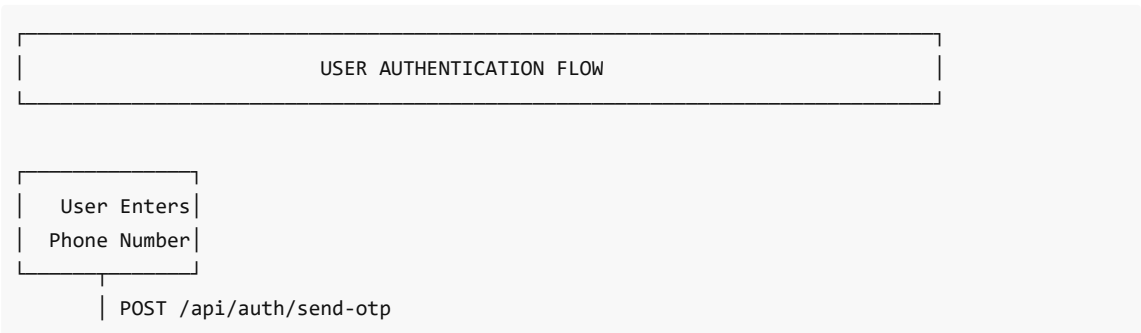


Discount Lala - Project Wireframes & Flow Diagrams

1. SYSTEM ARCHITECTURE OVERVIEW



2. USER AUTHENTICATION FLOW



Body: { phone: "+91xxxxxxxxx" }



Firestore sends OTP via SMS
Server returns sessionInfo



User Enters
OTP

POST /api/auth/verify-otp
Body: { sessionInfo, otp: "123456" }



Firestore verifies OTP
Returns: firebaseIdToken



POST /api/auth/verify-id-token
Body: { idToken: "firebase-id-token" }



User EXISTS
in DB

User NEW
(isNewUser)

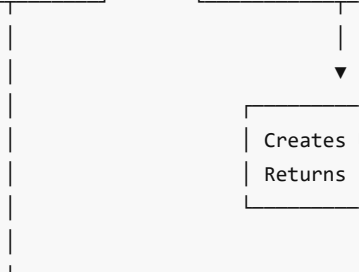


Returns JWT
token directly

POST /api/auth/register
Body: { firebaseIdToken
email?, name? }



Creates User in DB
Returns JWT token



AUTHENTICATED
JWT in Headers

for all requests

3. USER JOURNEY - BROWSE & PURCHASE FLOW

USER: BROWSE → CART → ORDER → COUPONS

STEP 1: BROWSE CONTENT (Public - No Auth Required)

GET /api/booklets
/city/:city_id

Returns:

- Booklet details
- Categories
- Linked Offers

GET /api/add-ons
/city/:city_id

Returns:

- Add-on details
- Categories
- Linked Offers

GET /api/booklets
/filter

Query: city,
price range,
status, validity

GET /api/add-ons
/filter

Query: city,
price range,
status, category

GET /api/banners
(public homepage)

(if route exists - banners displayed on home)

▼
User selects items

STEP 2: ADD TO CART (Auth Required - JWT Bearer Token)

POST /api/cart/add

Headers: Authorization: Bearer <JWT>

Body: { itemType: "booklet"|"add_on"|"coupon", itemId: uuid }

Auto-creates cart if user doesn't have one

```
GET /api/cart  
(view cart)
```

```
DELETE /api/cart/  
remove/:item_id
```

```
DELETE /api/cart/  
clear
```

|
| User proceeds to checkout
▼

STEP 3: CREATE ORDER (Auth Required)

```
POST /api/orders  
Headers: Authorization: Bearer <JWT>  
Body: {  
  distributorCode?: string, // Optional  
  referralCode?: string    // Optional  
}
```

Creates order from cart items + clears cart
Links distributor if code provided
Records referral if code provided

Order Status: "pending"

|
| User makes payment
▼

STEP 4: PAYMENT (Public - Payment Gateway Callback or Manual)

```
POST /api/payments  
Body: {  
  orderId: uuid,  
  amount: decimal,  
  paymentMethod: string,  
  paymentStatus: "success"|"failed"|"pending"  
}
```

IF paymentStatus == "success":

```
AUTO-TRIGGERS:  
1. Order status → "completed"  
2. Coupon generation for each order item
```

3. Distributor commission calculation (if applicable)

PUT /api/payments/
:id/status
(update status)

GET /api/payments/
order/:order_id
(view payments)

Payment success

STEP 5: COUPON GENERATION (Automatic)

For each OrderItem:

IF itemType == "booklet":
→ Fetch all BookletOffer links
→ Create UserCoupon for each offer
→ Expiry = booklet.validTo

IF itemType == "add_on":
→ Fetch all AddOnOffer links
→ Create UserCoupon for each offer
→ Expiry = offer.validTo

IF itemType == "coupon":
→ Create UserCoupon for the offer itself
→ Expiry = offer.validTo

IF order has distributorId:

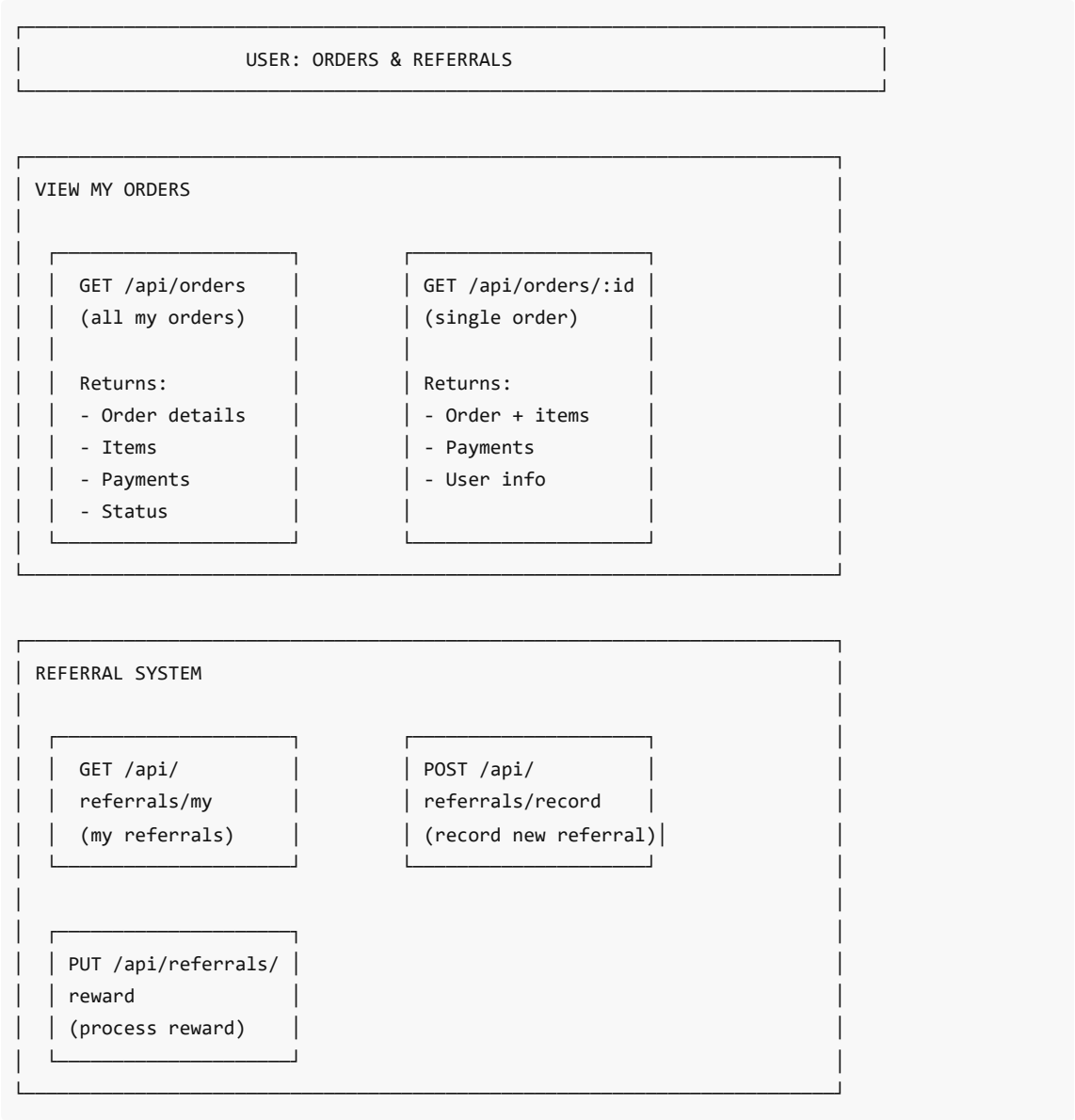
Create DistributorCommission:
amount = totalAmount * (distributor.commissionPercentage / 100)
status = "pending"

STEP 6: USER USES COUPONS (Auth Required)

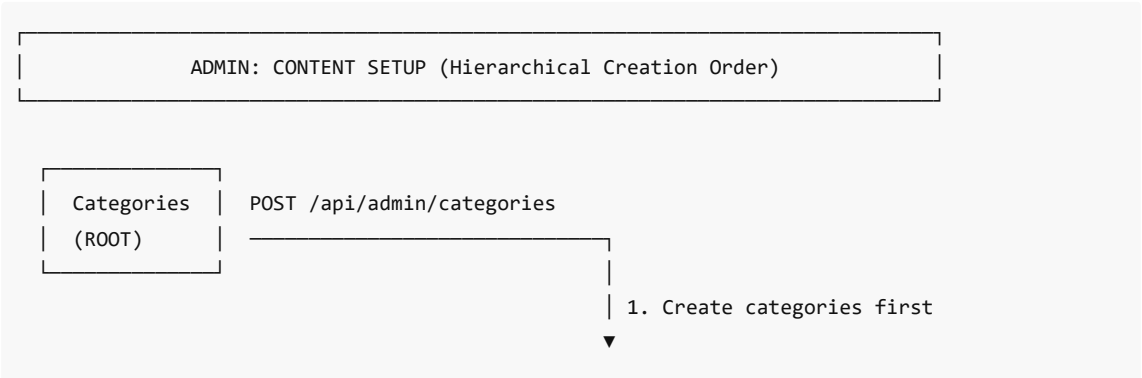
GET /api/coupons
Query: status?
(view my coupons)

POST /api/coupons/
:coupon_id/redeem
(mark as redeemed)

4. USER ORDER HISTORY & REFERRAL FLOW



5. ADMIN CONTENT MANAGEMENT FLOW



Places (under Cat)	POST /api/admin/places ← categoryId	GET all categories
-----------------------	--	-----------------------



Offers (at Place)	POST /api/admin/ offers ← placeId	GET all places
----------------------	--------------------------------------	-------------------

2. Create offers, then link to booklets or add-ons

Link Offers to Booklets: POST /api/admin/offers/booklet/add Body: { bookletId, offerId } (only offerType='booklet')		
Link Offers to Add-Ons: POST /api/admin/add-ons/add-offer Body: { addOnId, offerId } (only offerType='add-on')		



Booklets (1 per City)	POST /api/admin/ booklets ← cityId	(One per city max)
--------------------------	---------------------------------------	-----------------------



Add-Ons (per City)	POST /api/admin/ add-ons ← cityId	GET all cities
-----------------------	--------------------------------------	-------------------

ADMIN: CITIES (Top-Level Entity)		
POST	/api/admin/cities	Create city
GET	/api/admin/cities	List all cities
GET	/api/admin/cities/:id	Get city + its booklet
PUT	/api/admin/cities/:id	Update city
DELETE	/api/admin/cities/:id	Delete city (cascades to booklet)

ADMIN: BANNERS (Independent - for homepage display)		
POST	/api/admin/banners	Create banner (image, redirect)
GET	/api/admin/banners	List (ordered by priority DESC)
GET	/api/admin/banners/:id	Get single banner

PUT	/api/admin/banners/:id	Update banner
DELETE	/api/admin/banners/:id	Delete banner

6. ADMIN ORDER & DISTRIBUTOR MANAGEMENT

ADMIN: ORDER MANAGEMENT

GET /api/admin/orders

Returns ALL orders with:

- Order items
- Payments
- User info
- Distributor (if linked)

Admin updates order status

PUT /api/admin/orders/:id/status

Body: { status: "pending"|"processing"|"completed"|"cancelled" }

IF status == "completed":

- └ Generate UserCoupons for each order item
- └ Create DistributorCommission (if distributor linked)

ADMIN: DISTRIBUTOR MANAGEMENT

POST /api/admin/
distributors

Body: {
name, phone,
referralCode,
commissionPercent
}

GET /api/admin/
distributors

Returns all
distributors

GET /api/admin/

PUT /api/admin/

distributors/:id	distributors/:id
Returns distributor + orders + commissions	Update details

```
GET /api/admin/
distributors/
commissions

Query: distributorId?,
status?
```

7. ADMIN COUPON & REFERRAL MANAGEMENT

ADMIN: COUPON MANAGEMENT

```
GET /api/coupons/
admin/all

View ALL user
coupons across
all users
```

```
POST /api/coupons/
admin/create

Manually assign
coupon (offer) to
a user
```

```
POST /api/coupons/
admin/:id

Update coupon
status (active/
redeemed/expired)
```

ADMIN: REFERRAL MANAGEMENT

```
GET /api/referrals/
admin/all

View ALL referrals
```

across all users

8. COMPLETE API ROUTE MAP

ALL API ENDPOINTS

PUBLIC ENDPOINTS (No Auth Required)

AUTHENTICATION

POST	/api/auth/send-otp	Send Firebase OTP to phone
POST	/api/auth/verify-otp	Verify OTP, get Firebase token
POST	/api/auth/verify-id-token	Verify Firebase token → JWT or new
POST	/api/auth/register	Register new user → JWT

BROWSE (User-facing)

GET	/api/booklets/city/:city_id	Get booklets for a city
GET	/api/booklets/filter	Filter booklets
GET	/api/add-ons/city/:city_id	Get add-ons for a city
GET	/api/add-ons/filter	Filter add-ons

PAYMENTS

POST	/api/payments	Record payment for order
PUT	/api/payments/:id/status	Update payment status
GET	/api/payments/order/:order_id	Get payments for order

REFERRALS

POST	/api/referrals/record	Record a referral
PUT	/api/referrals/reward	Process referral reward

AUTHENTICATED ENDPOINTS (JWT Bearer Token Required)

CART

GET	/api/cart	Get user's cart
POST	/api/cart/add	Add item to cart
DELETE	/api/cart/remove/:item_id	Remove item from cart
DELETE	/api/cart/clear	Clear entire cart

ORDERS (User)

POST	/api/orders	Create order from cart
GET	/api/orders	Get user's orders
GET	/api/orders/:id	Get single order (owner/admin)

COUPONS (User)

GET	/api/coupons	Get user's coupons
POST	/api/coupons/:id/redeem	Redeem a coupon

REFERRALS (User)

GET	/api/referrals/my	Get user's referrals
-----	-------------------	----------------------

ADMIN COUPONS

GET	/api/coupons/admin/all	All user coupons
POST	/api/coupons/admin/create	Assign coupon to user
POST	/api/coupons/admin/:id	Update coupon status

ADMIN REFERRALS

GET	/api/referrals/admin/all	All referrals
-----	--------------------------	---------------

ADMIN ENDPOINTS (Currently Unprotected - No middleware)

CITIES

POST	/api/admin/cities	Create city
GET	/api/admin/cities	List cities
GET	/api/admin/cities/:id	Get city + booklet
PUT	/api/admin/cities/:id	Update city
DELETE	/api/admin/cities/:id	Delete city

CATEGORIES

POST	/api/admin/categories	Create category
GET	/api/admin/categories	List categories
GET	/api/admin/categories/:id	Get category + offers
PUT	/api/admin/categories/:id	Update category
DELETE	/api/admin/categories/:id	Delete category

PLACES

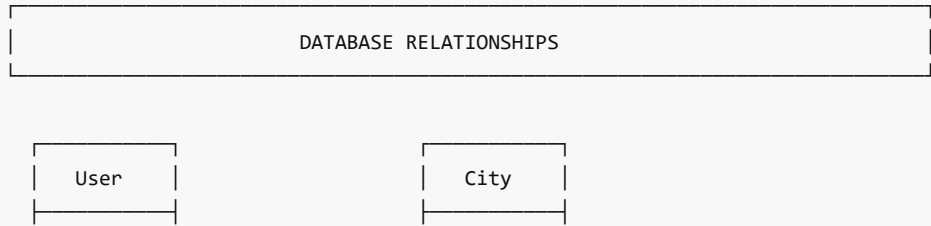
POST	/api/admin/places	Create place
GET	/api/admin/places	List places
GET	/api/admin/places/category/:id	Places by category
GET	/api/admin/places/:id	Get place + offers
PUT	/api/admin/places/:id	Update place
DELETE	/api/admin/places/:id	Delete place

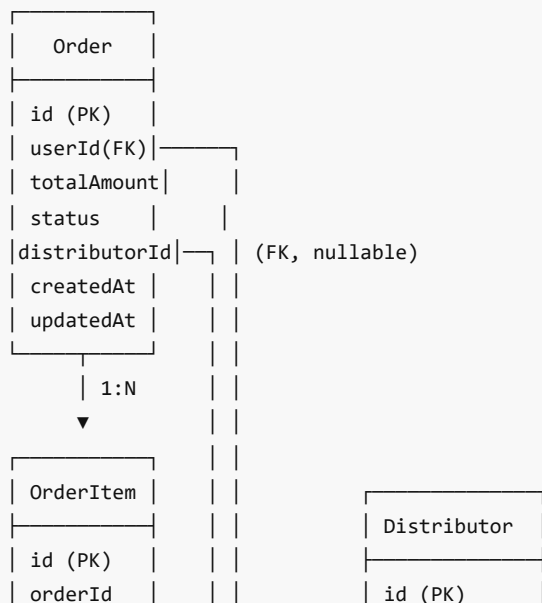
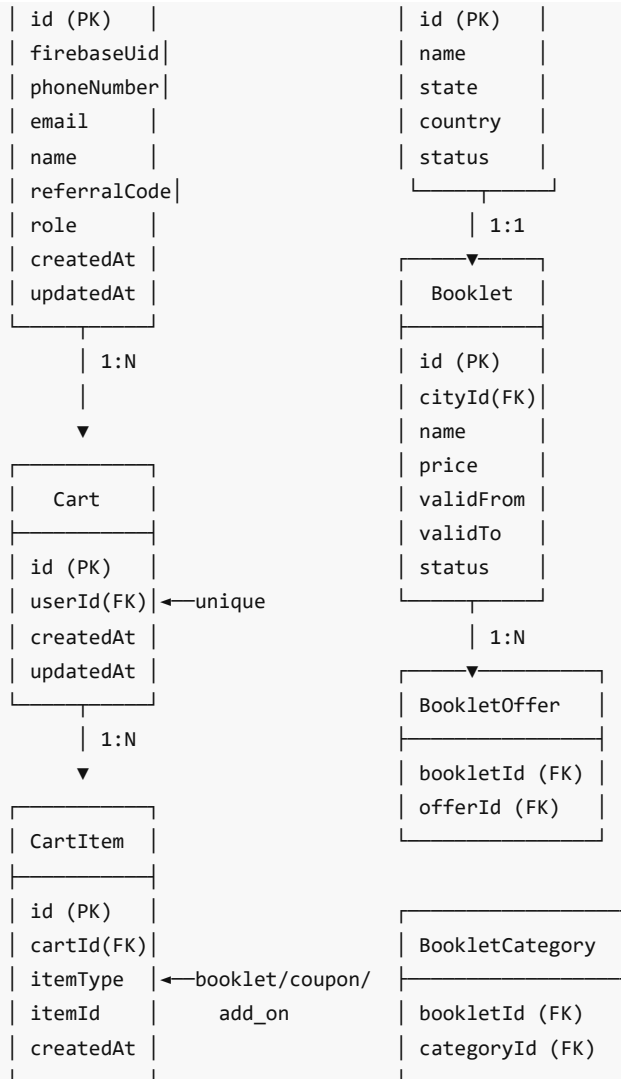
OFFERS

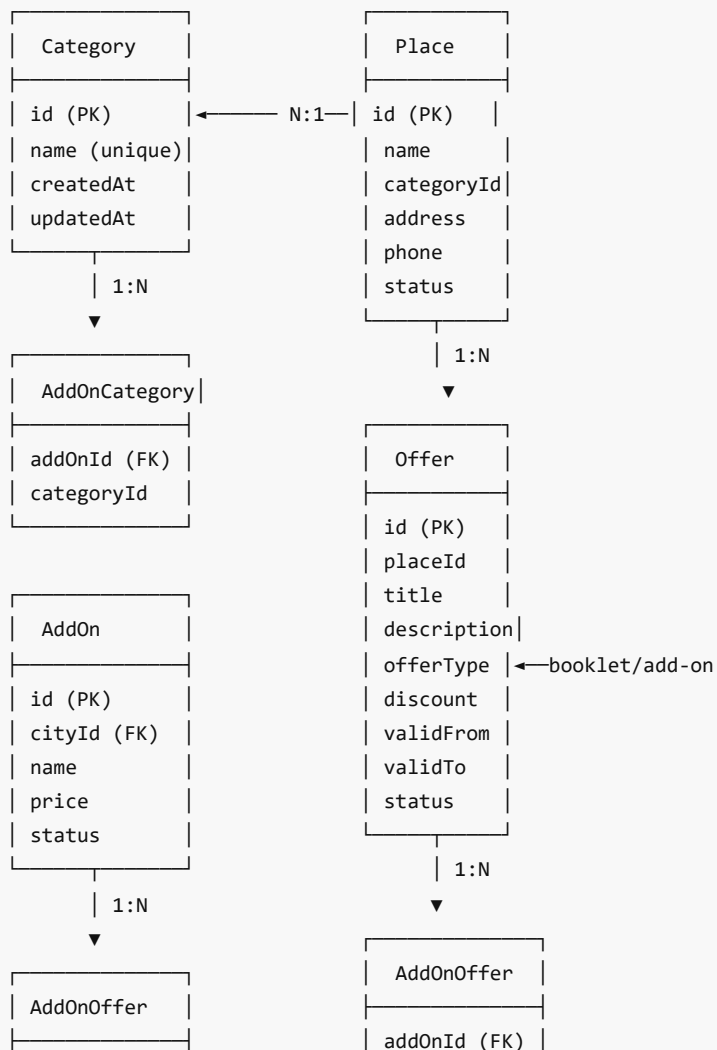
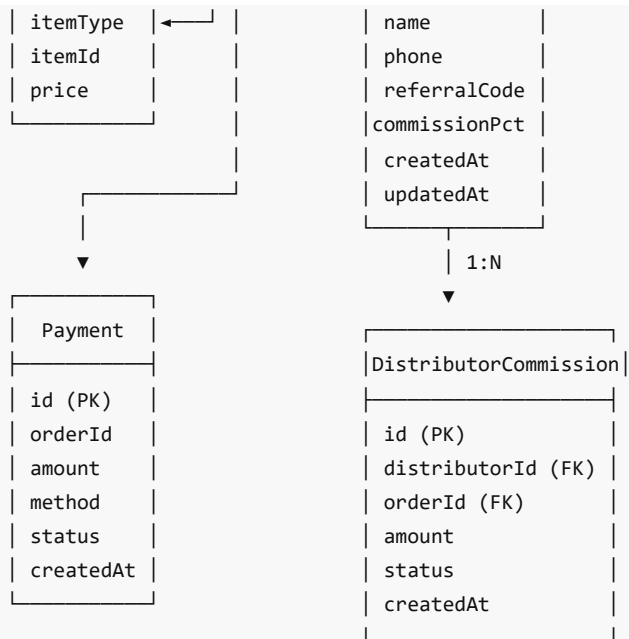
POST	/api/admin/offers	Create offer
GET	/api/admin/offers	List offers
GET	/api/admin/offers/:id	Get offer + links
PUT	/api/admin/offers/:id	Update offer
DELETE	/api/admin/offers/:id	Delete offer
POST	/api/admin/offers/booklet/add	Link offer to booklet
DELETE	/api/admin/offers/booklet/:bid	Unlink offer from booklet

/offer/:oid		
BOOKLETS		
POST	/api/admin/booklets	Create booklet
GET	/api/admin/booklets	List booklets
GET	/api/admin/booklets/:id	Get booklet
GET	/api/admin/booklets/:city_id	Get by city
PUT	/api/admin/booklets/:id	Update booklet
DELETE	/api/admin/booklets/:id	Delete booklet
ADD-ONS		
POST	/api/admin/add-ons	Create add-on
GET	/api/admin/add-ons	List add-ons
GET	/api/admin/add-ons/:id	Get add-on + links
PUT	/api/admin/add-ons/:id	Update add-on
DELETE	/api/admin/add-ons/:id	Delete add-on
POST	/api/admin/add-ons/add-offer	Link offer to add-on
DELETE	/api/admin/add-ons/:aid/ offer/:oid	Unlink offer from add-on
BANNERS		
POST	/api/admin/banners	Create banner
GET	/api/admin/banners	List banners (by priority)
GET	/api/admin/banners/:id	Get banner
PUT	/api/admin/banners/:id	Update banner
DELETE	/api/admin/banners/:id	Delete banner
ORDERS (Admin)		
GET	/api/admin/orders	All orders
PUT	/api/admin/orders/:id/status	Update order status
DISTRIBUTORS		
POST	/api/admin/distributors	Create distributor
GET	/api/admin/distributors	List distributors
GET	/api/admin/distributors/:id	Get distributor + orders
PUT	/api/admin/distributors/:id	Update distributor
GET	/api/admin/distributors/ commissions	List commissions

9. DATABASE ENTITY RELATIONSHIP DIAGRAM







addOnId (FK)	offerId (FK)
offerId (FK)	

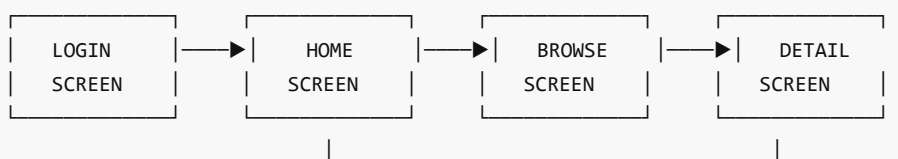
UserCoupon
id (PK)
userId (FK) → User
offerId (FK) → Offer
status → active/redeemed/expired
validFrom
validTo
redeemedAt

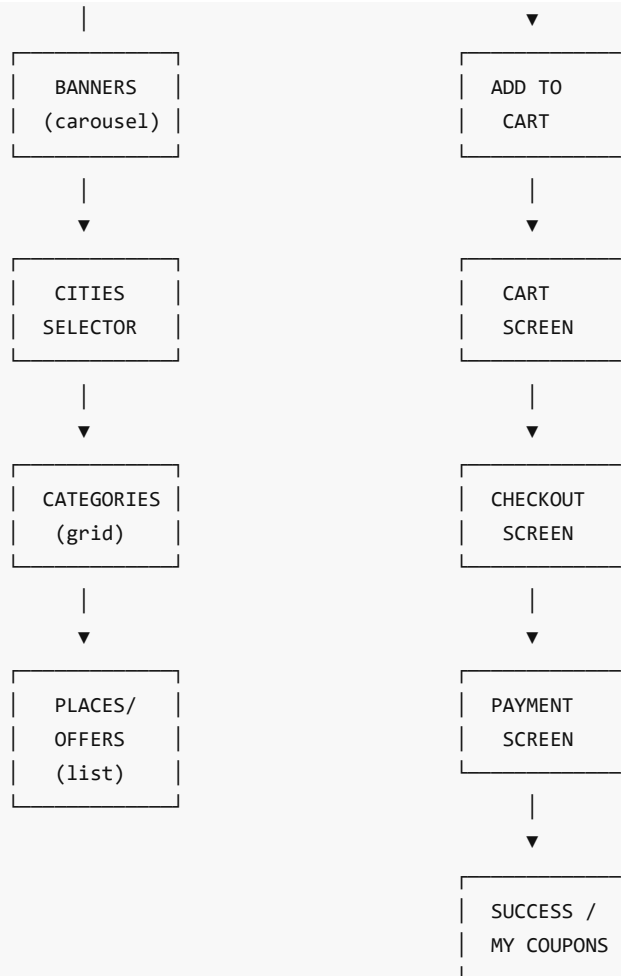
Banner
id (PK)
imageUrl
redirectType
redirectId
priority
isActive

ReferralLog
id (PK)
referrerId → User
referredId → User
status
rewardedAt
createdAt

10. SCREEN-BY-SCREEN WIREFRAME (USER APP)

USER APP SCREEN FLOW





SCREEN: LOGIN SCREEN
API: POST /api/auth/send-otp, /verify-otp, /verify-id-token, /register

Discount Lala
[Logo]

Phone Number: [+91 _____]

[SEND OTP]

_____ or _____

Enter OTP: [_ _ _ _ _]

[VERIFY]

(If new user)

Name: [_____]

Email: [] (optional)
Referral Code: [] (optional)
[COMPLETE REGISTRATION]

SCREEN: HOME SCREEN

API: GET /api/banners, City selector

[≡] Discount Lala [🔍] [🛒 Cart] [👤 Profile]

City Selector
📍 Current City: [Mumbai ▼]

BANNER CAROUSEL
[Banner 1] [Banner 2] [Banner 3]
◀ ● ● ● ▶

Browse by Category

🍔 Food | 🛍️ Shop | 🧑‍🦰 Beau | 🎬 Ent | (Category icons grid)
🏥 Health | ✈️ Travel | 🎓 Edu | 🏋️ Gym

Featured Booklets

Booklet Card	Booklet Card
[Image]	[Image]
Name, Price,	Name, Price,
Offers count	Offers count
[View Details]	[View Details]

Add-On Packages

Add-On Card	Add-On Card
Name, Price	Name, Price
[View Details]	[View Details]

SCREEN: BOOKLET / ADD-ON DETAIL SCREEN

API: GET /api/booklets/city/:id or GET /api/add-ons/city/:id

[← Back] Booklet Name

[🛒 Add to Cart]

Booklet Image

Price: ₹XXX

Validity: DD/MM - DD/MM

Description

Lorem ipsum dolor sit amet...

Categories Included

Food

Shop

Beauty

Offers Inside (12)



Restaurant Name



Address, Phone



20% OFF on all items



Valid till DD/MM/YYYY



Shop Name

...

[ADD TO CART - ₹XXX]

SCREEN: CART SCREEN

API: GET /api/cart, DELETE /api/cart/remove/:id, DELETE /api/cart/clear

[← Back] My Cart

[🗑️ Clear]



Mumbai Mega Booklet

₹299

[X Remove]



Premium Add-On Package

₹149

[X Remove]

Referral Code (optional)

[Enter code _____] [Apply]

Subtotal: ₹448

Total: ₹448

PROCEED TO CHECKOUT

SCREEN: CHECKOUT / PAYMENT SCREEN

API: POST /api/orders, POST /api/payments

[← Back] Checkout

Order Summary

2 items

₹448.00

Distributor Code (optional)

[Enter distributor code _____]

Payment Method

- ☐ UPI
- ☐ Credit/Debit Card
- ☐ Net Banking
- ☐ Wallet

PAY ₹448.00 & PLACE ORDER

After Payment Success:

✓ Payment Successful!

Your coupons are now available!

VIEW MY COUPONS

SCREEN: MY COUPONS SCREEN

API: GET /api/coupons, POST /api/coupons/:id/redeem

[≡] My Coupons

[Filter ▼]

[Profile]

Active (5) | Redeemed (2) | Expired (1)

20% OFF

Restaurant Name

📍 Address

Valid till: DD/MM/YYYY

Coupon Code: DISC-XXXXX

[REDEEM NOW]

Flat ₹100 OFF

Shop Name

...

[REDEEM NOW]

SCREEN: MY ORDERS SCREEN

API: GET /api/orders

[≡] My Orders

[Profile]

Order #ORD-XXXXX

Status: Completed ✓

Date: DD/MM/YYYY

2 items

₹448.00

Payment: Success

[View Details]

Order #ORD-XXXXX

Status: Pending ⌚

...

SCREEN: PROFILE SCREEN

API: User info, referrals

[← Back] My Profile



User Name

+91-XXXXXXXXXX



My Orders



My Coupons



My Referrals



My Referral Code: ABC123XYZ

[Copy]



Contact Support



Logout

11. SCREEN-BY-SCREEN WIREFRAME (ADMIN PANEL)

ADMIN PANEL SCREEN FLOW

ADMIN
LOGIN

DASHBOARD
(Stats)

MANAGE
CONTENT

CITIES
&BOOKLETS

CATEGOR
IES

PLACES

OFFERS

BANNERS

ADD-ONS

DISTRIBUT

ORDERS

COUPONS

SCREEN: ADMIN DASHBOARD

Admin Panel

[ Admin] [Logout]

Total Users	Total Orders	Revenue This Mo.	Active Coupons
1,234	567	₹45,678	890

Recent Orders

ORD-001	User X	₹299	Completed
ORD-002	User Y	₹448	Pending
ORD-003	User Z	₹149	Completed

Recent Referrals

User A → User B	Rewarded
User C → User D	Pending

SCREEN: ADMIN - CITY & BOOKLET MANAGEMENT

API: /api/admin/cities/*, /api/admin/booklets/*

Cities & Booklets Management

[+ Add City]

City Name	State	Status	Booklet	Actions
Mumbai	MH	Active	Mega Pack	[Edit][Delete]
Delhi	DL	Active	Delhi Deal	[Edit][Delete]
Bangalore	KA	Inactive	-	[Edit][Delete]

Add/Edit City

City Name:

State:

Country:

Status: [Active ▼]

Booklet (1 per city)

Booklet Name:

Price:

Valid From: [DD/MM/YYYY]

Valid To: [DD/MM/YYYY]

Categories: ☒ Food ☒ Shop ☐ Beauty

[Save] [Cancel]

SCREEN: ADMIN - CATEGORY & PLACE MANAGEMENT

API: /api/admin/categories/*, /api/admin/places/*

Categories Management [+ Add Category]

Category	Places Count	Offers Count	Actions
Food	45	120	[Edit][Delete]
Shopping	32	89	[Edit][Delete]
Beauty	18	45	[Edit][Delete]

Places Management (Category: Food) [+ Add Place]

Place Name	Address	Phone	Status	Actions
KFC	Andheri W	98xxx	Active	[Edit][Delete]
McDonald's	Bandra	97xxx	Active	[Edit][Delete]

Add/Edit Place

Name:

Category: [Food ▼]

Address:

Phone:

Status: [Active ▼]

[Save] [Cancel]

SCREEN: ADMIN - OFFER MANAGEMENT

API: /api/admin/offers/*

Offers Management

[+ Add Offer]

Filter: [All Places ▼] [All Status ▼]

Title	Place	Type	Valid Till	Actions
20% OFF	KFC	booklet	31/12/2025	[Edit][Del]
Flat ₹100	Zara	add-on	30/06/2025	[Edit][Del]

└ Add/Edit Offer

Title: [_____]

Description: [_____]

Place: [KFC ▼]

Offer Type: [booklet ▼] (booklet / add-on)

Discount: [_____]

Valid From: [DD/MM/YYYY]

Valid To: [DD/MM/YYYY]

Status: [Active ▼]

[Save] [Cancel]

Link Offers to Booklets/Add-Ons

[Link to Booklet] Booklet: [Mega Pack ▼] Offer: [20% OFF ▼]

[Add]

[Link to Add-On] Add-On: [Premium ▼] Offer: [Flat ₹100 ▼]

[Add]

SCREEN: ADMIN - ADD-ON MANAGEMENT

API: /api/admin/add-ons/*

Add-On Packages Management

[+ Add Add-On]

Filter: [All Cities ▼] [All Status ▼]

Name	City	Price	Offers	Status	Actions
Premium Pkg	Mumbai	₹149	8	Active	[E][D]
Basic Pkg	Delhi	₹99	5	Active	[E][D]

Add/Edit Add-On

Name:

City:

[Mumbai ▼]

Price:

Status:

[Active ▼]

Categories:

☒ Food

☒ Shopping

☐ Beauty

[Save]

[Cancel]

SCREEN: ADMIN - ORDER MANAGEMENT

API: /api/admin/orders/*

Order Management

Filter: [All Status ▼] [Search Orders...]

Order ID	User	Amount	Status	Actions
ORD-001	+91xxxxx	₹299.00	Pending	[View]
ORD-002	+91yyyyy	₹448.00	Completed	[View]
ORD-003	+91zzzzz	₹149.00	Cancelled	[View]

Order Detail View

Order: ORD-001

User: +91-XXXXXX XXXXXX

Date: DD/MM/YYYY HH:MM

Items:

Mumbai Mega Booklet

₹299.00

Premium Add-On

₹149.00

Total: ₹448.00

Payments:

UPI | ₹448.00 | Success | DD/MM/YYYY

Distributor: DIST-001 (if applicable)

Update Status: [Completed ▼] [Update]

SCREEN: ADMIN - DISTRIBUTOR MANAGEMENT

API: /api/admin/distributors/*

Distributor Management

[+ Add Distributor]

Name	Phone	Code	Comm%	Orders	Actions
John D	+91xxxxx	JD001	10%	25	[E][D]
Jane S	+91yyyyy	JS002	15%	12	[E][D]

└ Add/Edit Distributor

Name: [_____]

Phone: [_____]

Referral Code: [_____]

Commission %: [____]

[Save] [Cancel]

└ Commission Reports

Filter: [All Distributors ▼] [All Status ▼]

Distributor	Order	Amount	Status	Date
John D	ORD-001	₹29.90	Pending	...
John D	ORD-005	₹44.80	Paid	...
Jane S	ORD-003	₹22.35	Pending	...

SCREEN: ADMIN - BANNER MANAGEMENT

API: /api/admin/banners/*

Banner Management

[+ Add Banner]

Priority	Image	Redirect Type	Redirect ID	Active
3	[img]	booklet	uuid-xxx	✓
2	[img]	add-on	uuid-yyy	✓
1	[img]	external	url	X

└ Add/Edit Banner

Image: [Upload Image] [Current: img.jpg]

Redirect Type: [booklet ▼] (booklet / add-on / external)

Redirect ID: [_____]

Priority: [____] (higher = shown first)

Active: ☒

[Save] [Cancel]

12. CONTENT CREATION WORKFLOW (Admin Setup Sequence)

ADMIN CONTENT SETUP - STEP BY STEP SEQUENCE

STEP 1: CREATE CATEGORIES

POST /api/admin/categories

→ Food, Shopping, Beauty, Entertainment, Health, Travel, etc.



STEP 2: CREATE PLACES (under categories)

POST /api/admin/places (with categoryId)

→ KFC, McDonald's, Zara, etc.



STEP 3: CREATE OFFERS (at places)

POST /api/admin/offers (with placeId, offerType)

→ "20% OFF", "Flat ₹100", etc.

→ offerType: "booklet" or "add-on"

STEP 4A: CREATE CITIES

```
POST /api/admin/cities
→ Mumbai, Delhi, Bangalore
```

STEP 4B: CREATE DISTRIBUTORS

```
POST /api/admin/distributors
→ Name, phone, code, comm%
```

STEP 5: CREATE BOOKLETS (1 per city)

```
POST /api/admin/booklets (with cityId)
→ Name, price, validity, categories
```

STEP 6: LINK OFFERS TO BOOKLETS

```
POST /api/admin/offers/booklet/add
→ Link offers (offerType="booklet") to booklets
```

STEP 7: CREATE ADD-ONS (per city)

```
POST /api/admin/add-ons (with cityId)
→ Name, price, categories
```

STEP 8: LINK OFFERS TO ADD-ONS

```
POST /api/admin/add-ons/add-offer
→ Link offers (offerType="add-on") to add-ons
```

STEP 9: CREATE BANNERS (optional)

```
POST /api/admin/banners
→ Image, redirect, priority
```

RESULT: Users can now browse booklets/add-ons, add to cart, purchase,
and receive coupons (offers) that they can redeem at places

13. KEY OBSERVATIONS & RECOMMENDATIONS

Security Issues Found:

1. **Admin routes are unprotected** - No `authMiddleware` on `/api/admin/*` routes
2. **Role checks are manual** - Some controllers check `req.user.role` but admin routes bypass auth middleware entirely
3. **OTP sessions are in-memory** - Will be lost on server restart

Flow Gaps:

1. No public endpoint to fetch banners (only admin CRUD exists)
2. No user-facing endpoint to view places directly
3. Referral reward processing is manual (no automatic trigger on order completion)

Recommendations:

1. Add `authMiddleware` + role check to all admin routes
2. Add public GET endpoint for banners
3. Consider auto-processing referral rewards when referred user completes first order
4. Add pagination to list endpoints (currently returns all records)
5. Add rate limiting on OTP endpoints